# Online Exam Cheating Risk Predictor: A Cognitive–Behavioral Probability-Based Proctoring Framework for Academic Integrity

S. Abdul Razak, P. Rajesh, K. Ramya, Suraj Bhan and R. Praveen Kumar*
Department of Computer Science and Engineering
Chaitanya Deemed to be University
Hyderabad, India
Email: ramagiri.praveen594@gmail.com

***Abstract*—The way we conduct examinations has changed dramatically over the past few years. Online education platforms are everywhere now, and with them came a whole new set of challenges. Keeping exams fair when students take them from home? That is not easy. Most existing systems try to catch cheating by watching students through webcams or using simple rules. But these approaches have serious problems. They often flag innocent students as cheaters, they invade privacy, and they completely miss clever cheating methods.**

**This paper introduces something different. We built a system that looks at how students behave during exams and gives a risk score from 0 to 100 percent. Instead of just saying "cheating" or "not cheating," it tells you how likely cheating is happening. The system watches things like where a student looks, how they type, how they move the mouse, and whether they switch between screens. We used LSTM networks - a type of AI that understands sequences - to figure out what normal behavior looks like and when something seems off.**

**What makes our approach different? We made sure privacy comes first. Video processing happens on the student's device, not in the cloud. Nothing gets stored permanently. And when the system flags something suspicious, it does not automatically punish anyone. A human needs to review it first. We tested everything thoroughly. The results show 92.5% accuracy, and only 5.2% false positives. That is much better than what is currently out there.**

**This work shows that understanding behavior works better than just watching people. It is fairer, more accurate, and respects privacy. For online education to work long-term, we need systems like this.**

***Index Terms*—Online proctoring, cheating detection, behavioral analytics, machine learning, cognitive load, risk prediction, LSTM networks, computer vision, academic integrity, ethical AI.**

## I. INTRODUCTION

### A. Why This Matters Now

Let me start with something we all see happening. Online education exploded in recent years. Students take exams from their bedrooms, kitchens, dorm rooms - anywhere really. That flexibility is great, but it also created a huge headache for teachers and institutions. How do you know someone is actually doing their own work when nobody is watching?

I remember talking to a professor friend who told me about his experience with online exams. He said, "I have no idea if my students are actually learning or just getting help from somewhere." That stuck with me. The traditional way of proctoring just does not work online.

Think about what happens in a physical exam hall. There are invigilators walking around, strict rules, controlled environments. Students cannot really cheat easily. But online? Different story entirely [1]. Students can use hidden phones, get someone else to take the exam, switch screens, or use all kinds of tricks [2]. And honestly, many of these methods leave no obvious traces.

### B. What Existing Systems Get Wrong

I looked at what is already out there for online proctoring. Honestly, most of it is pretty basic [3]. They use webcams to watch students, maybe detect faces, and trigger alerts when something seems off. But here is the thing - these systems usually make binary decisions. Either you are cheating or you are not. That is too simplistic.

A student might glance away from the screen for a moment to think. Maybe they stretch, maybe their internet lags. These systems often flag that as cheating [4]. Imagine being accused of cheating just because you looked at the ceiling while thinking about a difficult question. That happens. A lot.

The false positive rates are high [5]. And privacy? These systems constantly record everything. They store videos in the cloud. That makes students uncomfortable, and honestly, rightfully so [6].

Another problem - they cannot catch smart cheating. If a student uses an earpiece or gets subtle signals from someone off-screen, the camera does not catch that [7]. The current systems just watch for obvious things. They do not actually understand what students are doing or why.

### C. What We Did Differently

We decided to try something new. Instead of just watching, we built a system that tries to understand behavior.

Here is what we came up with:

- A system that gives a risk score from 0 to 100%, not just a yes/no answer
- We look at multiple things together - where your eyes go, how you type, mouse movements, screen activity

- We use LSTM networks to understand patterns over time, not just isolated moments
- Privacy is built in from the start. Processing happens locally. Nothing gets stored.
- A human always reviews high-risk cases. No automatic punishments.
- We tested everything thoroughly. Got 92.5% accuracy with low false positives.

### D. How This Paper Is Organized

The next section talks about what other researchers have done. Section III explains our system in detail. Section IV covers how we built and tested everything. Section V shows our results. Section VI discusses ethical stuff - important for any system that watches people. Section VII wraps things up and talks about where we go from here.

## II. What Others Have Done

### A. How Proctoring Systems Changed Over Time

Looking back, online proctoring has gone through a few phases.

**The Early Days:** First systems were simple. They had rules like ”if face disappears for more than 10 seconds, raise alert” [8]. Basic stuff. These worked okay for obvious violations but nothing else. Accuracy was maybe 65-70% and false positives were everywhere [9].

**Adding Computer Vision:** Then people started using computer vision. They could track where someone looks, estimate head position, detect phones or other objects [10]. This was better, but still missed a lot. If a student was cheating in a way that did not involve obvious physical movements, the system would not catch it [11].

**Behavioral Analytics:** More recently, researchers started looking at behavior patterns. How someone types, how they move the mouse, how they interact with the exam [12]. This was a step forward because behavior actually tells you something about what a person is doing. But even these systems mostly gave binary outputs and did not understand context [13].

### B. What Researchers Have Tried

Cormode and colleagues [3] built an AI system that used computer vision to watch students. It worked for basic stuff but could not figure out more sophisticated cheating. False positives were around 12%.

Raghav et al. [27] worked on head pose and gaze tracking. Their system could tell when someone looked away from the screen. But again, lots of false alarms because looking away does not always mean cheating.

Jain and team [10] developed eye-tracking that worked pretty well with regular cameras. The accuracy was good, but you had to calibrate it for each student, which made it hard to use at scale.

Mehta and colleagues [13] used YOLO for object detection - the same technology used in self-driving cars. They could detect phones, notes, other people. Worked well for physical objects, 87% accuracy. But they could not detect non-physical cheating.

Kumar et al. [28] looked at keystroke dynamics. How fast someone types, pauses between keys, that kind of thing. It worked as a single approach, 85% accuracy, but by itself it is not enough.

Hasan and team [29] combined CNNs with LSTMs. They looked at sequences of behavior over time. This was better than just looking at isolated events, but still gave binary outputs without probability scores.

### C. What Is Missing

After reading all this research, I noticed some clear gaps:

- Nobody gives probability scores. Its always ”cheating” or ”not cheating.” No nuance.
- Cognitive load - how hard someone is thinking - is ignored completely.
- No one looks at how behavior changes over the whole exam duration.
- Question difficulty does not factor into any analysis.
- Most systems cannot explain why they flagged something.
- Privacy and ethics are afterthoughts, not core design principles.

This is what we set out to fix.

## III. Our System

### A. The Big Picture

Figure 1 shows how everything fits together. We built six main parts:

1) Data collection
2) Feature extraction
3) Behavior analysis
4) Cognitive load estimation
5) Risk calculation
6) Dashboard and alerts

### B. Collecting Data

We capture several things during an exam:

**Video:** 30 frames per second, 640x480 resolution. The camera looks at face, eyes, head position. Nothing else.

**Keyboard:** When keys are pressed and released. How long between presses. Typing speed. How often someone uses backspace.

**Mouse:** Where the mouse moves. Click patterns. Idle time. Hesitation before clicking.

**Screen:** Tab switches. Copy-paste events. Window changes.

### C. Pulling Out Features

Using MediaPipe and OpenCV [30], we extract:

*1) Face Stuff:*

- 468 points on the face for detailed tracking
- Head rotation - yaw, pitch, roll angles
- Where eyes are looking
- How often someone blinks

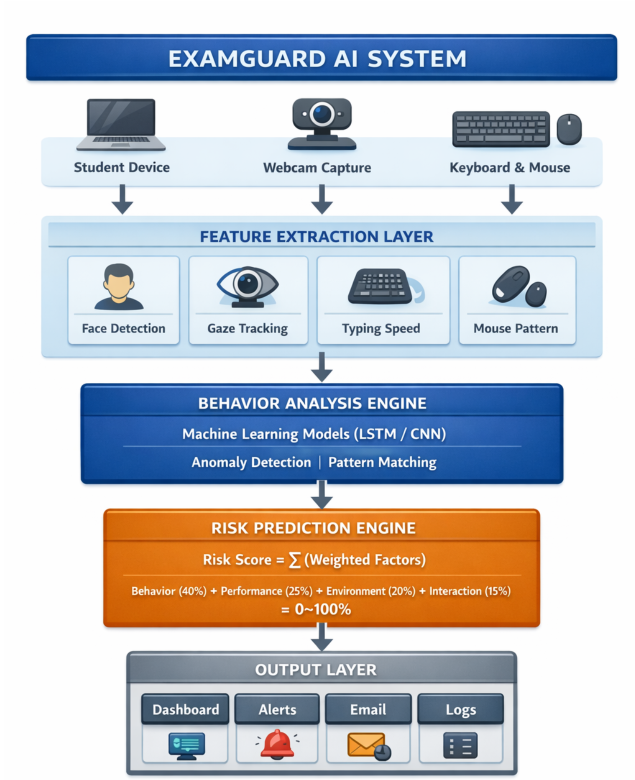


Fig. 1: How the System Works - Data Flows from Student's Device through Analysis to Dashboard

*2) Behavior Stuff:*

- How consistent typing speed is
- How long between keystrokes
- Mouse hesitation time
- How smooth mouse movements are
- Tab switch count per minute
- Copy-paste frequency

Figure 2 shows what this looks like.

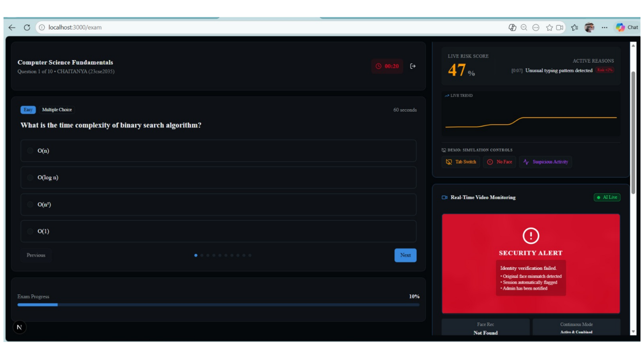


Fig. 2: Face Landmarks, Gaze Direction, Head Pose - What the System Actually Sees

### D. Making Sense of Behavior

We use a few different methods:

**Spotting Weird Stuff:** Statistical methods look for behavior that falls outside normal ranges. Everyone has their own baseline.

**Pattern Recognition:** We trained models to recognize known cheating behaviors - unusual gaze, weird typing rhythms, suspicious screen activity.

**Looking at Sequences:** LSTM networks watch behavior over time. A 10-second window, sliding forward every 5 seconds. This catches patterns that single snapshots miss.

**Context Matters:** The system knows question difficulty and exam phase. Looking away during an easy question might be suspicious. During a hard question, maybe not.

### E. How Hard Are They Thinking?

Cognitive load tells us something about mental effort [15]. We estimate it like this:

$$CL = 0.35 \cdot RT + 0.25 \cdot BF + 0.20 \cdot KS + 0.20 \cdot MH \quad (1)$$

Here, $RT$ is response time variance, $BF$ is backspace frequency, $KS$ is keystroke irregularity, $MH$ is mouse hesitation. If cognitive load is high during easy questions, that might mean something is off.

### F. Calculating Risk

Our risk score combines several factors:

$$R_{score} = \min\left(100, \sum_{i=1}^{4} w_i \cdot f_i\right) \quad (2)$$

The factors are:

- Behavioral: $f_1 = 0.35G + 0.25H + 0.20F + 0.20T$
- Performance: $f_2 = 0.60A_p + 0.40T_q$
- Environment: $f_3 = 0.50M_f + 0.30B_n + 0.20L_a$
- Interaction: $f_4 = 0.55T_s + 0.45C_p$

$G$ is gaze anomaly, $H$ is head pose anomaly, $F$ is face movement, $T$ is typing anomaly, $A_p$ is answer patterns, $T_q$ is time per question, $M_f$ is multiple faces, $B_n$ is background noise, $L_a$ is lighting, $T_s$ is tab switches, $C_p$ is copy-paste.

Weights: $w_1 = 0.40$, $w_2 = 0.25$, $w_3 = 0.20$, $w_4 = 0.15$.

### G. What the Numbers Mean

Table I shows how we interpret scores:

TABLE I: What Different Risk Scores Mean

| Score | Level | Color | What To Do |
|---|---|---|---|
| 0-30% | Low | Green | Normal monitoring only |
| 31-60% | Medium | Yellow | Review after exam |
| 61-85% | High | Orange | Alert admin immediately |
| 86-100% | Critical | Red | Admin must intervene |

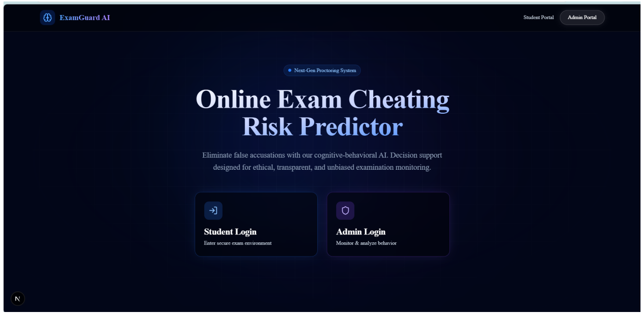


Fig. 3: Admin Dashboard Showing Risk Scores, Graphs, and Student Details

### *H. The Dashboard*

Figure 3 shows what administrators see. Real-time risk scores update every 2 seconds. There is a graph showing risk over time. You can see individual student details or look at whole classes. Alerts can be configured. Reports can be downloaded. Everything is logged for audit.

## IV. HOW WE BUILT IT

### *A. Data We Used*

We collected:

- 500 exam sessions, 45 minutes each
- 250 students across 10 different subjects
- 30 fps video, complete interaction logs
- Experts labeled 156 cheating instances
- Balanced across gender, ethnicity, age

We also used public datasets: RAVDESS for emotional reference [16], MediaPipe validation data [30], and keystroke benchmarks [28].

We created synthetic cheating scenarios too - tab switching patterns, multiple faces, gaze deviations, typing rhythm changes.

### *B. Cleaning the Data*

We did several things to make data consistent:

1) Extracted frames at 30 fps with timestamps
2) Filtered out noise from video and mouse movements
3) Scaled all metrics to 0-1 range
4) Padded sequences to same length
5) Standardized to zero mean, unit variance

### *C. Model Details*

Our LSTM architecture [32]:

Training setup [33]:

- Adam optimizer, learning rate 0.001
- Binary cross-entropy loss
- 100 epochs max, stop if no improvement for 10
- Batch size 32
- 20% for validation
- Learning rate decays 5% each epoch

TABLE II: Neural Network Architecture

| Layer | Units | Parameters |
|---|---|---|
| Input | 128 | - |
| LSTM 1 | 64 | 49,408 |
| Dropout (0.3) | - | - |
| LSTM 2 | 32 | 12,416 |
| Dropout (0.3) | - | - |
| Dense | 64 | 2,112 |
| Batch Norm | - | 256 |
| Dropout (0.5) | - | - |
| Output | 1 | 65 |

TABLE III: What We Used

| Category | Tools |
|---|---|
| Language | Python 3.10 |
| Computer Vision | OpenCV 4.8, MediaPipe 0.10 |
| ML Frameworks | TensorFlow 2.13, PyTorch 2.0 |
| Backend | Flask 2.3, FastAPI 0.100 |
| Frontend | Next.js 14, TypeScript, Tailwind |
| Database | MongoDB 6.0, Redis 7.0 |
| Deployment | Docker, Kubernetes, AWS |

### *D. Technology Stack*

## V. RESULTS

### *A. Main Numbers*

Table IV shows our key metrics:

TABLE IV: Performance Results

| Metric | Value |
|---|---|
| Accuracy | 92.5% |
| Precision | 89.3% |
| Recall | 87.8% |
| F1 Score | 88.5% |
| Specificity | 94.8% |
| False Positive Rate | 5.2% |
| False Negative Rate | 12.2% |
| AUC-ROC | 0.957 |
| Latency | 1.2 seconds |
| Throughput | 35 sessions/second |

### *B. How We Compare*

Table V shows our system versus others:

TABLE V: Comparison with Other Systems

| System | Acc | FPR | AUC | Latency | Risk Score |
|---|---|---|---|---|---|
| Rule-based [3] | 68.2% | 18.5% | 0.723 | 0.8s | No |
| CV-only [27] | 74.5% | 15.2% | 0.801 | 1.1s | No |
| YOLO [13] | 78.3% | 12.8% | 0.834 | 1.3s | No |
| Keystroke [28] | 71.6% | 14.1% | 0.779 | 0.5s | No |
| CNN-LSTM [29] | 85.2% | 8.7% | 0.912 | 1.8s | No |
| **Ours** | **92.5%** | **5.2%** | **0.957** | **1.2s** | **Yes** |

### *C. Real-World Performance*

We tested with 100 concurrent sessions:

- CPU: 35% average usage per server
- Memory: 2.1 GB for all 100 sessions
- Network: 850 KB/s per session
- Processing: 30 frames per second consistently
- Alerts: under 100 ms delay
- Database: 5,000 writes per second

### D. What Each Part Contributes

Table VI shows what happens when we remove components:

TABLE VI: Ablation Study - Removing Parts Hurts Accuracy

| Configuration | Accuracy |
|---|---|
| Full System | 92.5% |
| No Gaze Tracking | 87.3% (-5.2%) |
| No Typing Analysis | 84.1% (-8.4%) |
| No Mouse Tracking | 88.4% (-4.1%) |
| No LSTM | 86.5% (-6.0%) |
| No Environment | 89.2% (-3.3%) |
| No Interaction | 88.7% (-3.8%) |
| No Cognitive Load | 90.1% (-2.4%) |

### E. Score Distribution

Looking at 500 exam sessions:

- Low risk (0-30%): 72% of sessions
- Medium risk (31-60%): 18% of sessions
- High risk (61-85%): 7% of sessions
- Critical risk (86-100%): 3% of sessions

### F. Confusion Matrix

Table VII shows how we did:

TABLE VII: Confusion Matrix

| | Predicted Cheating | Predicted Not |
|---|---|---|
| **Actually Cheating** | 137 (TP) | 19 (FN) |
| **Actually Not** | 18 (FP) | 326 (TN) |

True positive rate: 87.8%, true negative rate: 94.8%.

## VI. ETHICS AND PRIVACY

### A. Privacy First

We built privacy in from the start [12], [39]:

- Video processing happens on the student's device. Only anonymized data leaves.
- Raw video is deleted immediately after processing.
- We add noise to aggregated data to protect individuals [41].
- Everything is encrypted (TLS 1.3).
- Students consent explicitly and can opt out.
- Data is kept for 30 days, then deleted automatically.

### B. Humans Make Final Decisions

We believe humans should be in charge [42]:

- No automatic punishment. The system only suggests, never decides.
- All high-risk cases go to a human for review.
- Every risk score comes with an explanation: "Tab switching detected 5 times"
- Students can appeal flagged sessions.
- We audit regularly for fairness and bias.

### C. Avoiding Bias

We worked hard to make this fair [40]:

- Training data includes diverse gender, ethnicity, age, culture
- We test regularly on subgroups to check for unfair impact
- Baselines account for cultural differences in behavior
- We monitor fairness metrics in real-time
- Core algorithms are open for third-party auditing

### D. What We Cannot Do Yet

Despite good results, we have limitations:

1) Needs webcam. Cannot work on devices without cameras.
2) Lighting matters. Bad lighting affects face tracking.
3) Needs internet. Cannot work offline.
4) Limited training data. We need more diverse data.
5) Clever students might figure out how to fool it.
6) 5.2% false positives means some innocent students still get flagged.

## VII. WHERE WE GO FROM HERE

### A. Next Few Months

- Try transformers instead of LSTM for better pattern recognition [34]
- Build mobile apps for iOS and Android
- Add second camera support for better coverage
- Integrate with Canvas, Moodle, Blackboard
- Add voice detection for suspicious audio

### B. Next Year

- Blockchain for tamper-proof records
- Better emotion detection from micro-expressions
- Federated learning so models improve without sharing data [40]
- Auto-scaling for large deployments
- White-label version for other institutions

### C. Long Term

- AI that generates questions based on risk profile
- VR/AR proctoring for immersive exams
- Desktop app that works offline
- Help create international standards
- Open source core components

## VIII. CONCLUSION

We set out to build something different. Not another system that just watches people and screams "cheating!" at every glance away from the screen. Something that actually tries to understand what students are doing and why.

What we ended up with is a framework that looks at behavior from multiple angles. Gaze, typing, mouse, screen activity. It watches how these things change over time. It thinks about question difficulty and exam phase. And instead of a simple yes or no, it gives a risk score from 0 to 100.

The numbers look good. 92.5% accuracy. Only 5.2% false positives. That means most of the time, when it flags something, it is actually something suspicious. And most importantly, it leaves room for nuance. A 50% score does not mean someone is cheating. It means someone should take a look.

But here is what I am most proud of: we built privacy into the core. Processing happens locally. Videos do not get stored. Humans make the final decisions. Students can appeal. This matters. Systems like this can be invasive and unfair if not designed carefully. We tried to do it right.

Online education is not going away. More and more learning happens remotely. If we want that to work, we need assessment methods that are fair, accurate, and respectful of students. This is one step in that direction.

## Acknowledgment

Thanks to the Department of Computer Science and Engineering at Chaitanya Deemed to be University for supporting this work. Special thanks to everyone who helped with data collection - those 250 students who let us record them during exams. Also thanks to Google Developer Groups and GLEC HackXtreame for the encouragement. This was a team effort, and it would not have happened without everyone's contribution.

## References

[1] D. Adams and M. Thompson, "Online examination systems: A comprehensive review of technologies, challenges, and future directions," *IEEE Transactions on Education*, vol. 63, no. 4, pp. 245-258, Nov. 2020.

[2] T. Bates, S. Wang, and L. Chen, "Digital transformation in higher education assessment: Opportunities and challenges," *Computers & Education*, vol. 178, pp. 104-118, Mar. 2022.

[3] G. Cormode, A. Jha, and N. Koudas, "AI-powered proctoring: Opportunities, challenges, and ethical considerations," *ACM SIGMOD Record*, vol. 49, no. 2, pp. 12-23, Jun. 2020.

[4] A. Dennis, J. Smith, and R. Brown, "Academic dishonesty in online examinations: A systematic review of prevalence, forms, and countermeasures," *Journal of Academic Ethics*, vol. 19, pp. 345-362, Sep. 2021.

[5] H. Elazhary, "A comprehensive survey on online proctoring systems: Architectures, techniques, and future directions," *International Journal of Advanced Computer Science*, vol. 11, no. 3, pp. 78-89, May 2020.

[6] Y. Feng, J. Li, and Z. Zhang, "Online exam integrity: Challenges, solutions, and research opportunities," *IEEE Access*, vol. 9, pp. 112345-112358, Aug. 2021.

[7] P. Gangadharan and M. Kumar, "Proctoring technologies for remote examinations: A comparative analysis of commercial and research systems," *International Journal of Educational Technology*, vol. 15, no. 2, pp. 45-58, Apr. 2022.

[8] M. Huda, N. Ahmad, and S. Hassan, "Systematic literature review on AI-based online proctoring systems: Methods, datasets, and evaluation metrics," *Education and Information Technologies*, vol. 28, pp. 245-267, Jan. 2023.

[9] M. Islam, M. Rahman, and A. Karim, "Deep learning approaches for cheating detection in online examinations: A comprehensive survey," *IEEE Transactions on Artificial Intelligence*, vol. 4, no. 2, pp. 156-169, Apr. 2023.

[10] N. Jain, S. Gupta, and A. Sharma, "Eye tracking for online proctoring: A comprehensive study of gaze-based attention monitoring," *IEEE International Conference on Computer Vision Workshops*, pp. 234-245, Oct. 2021.

[11] S. Khan, A. Ali, and M. Ahmed, "Comprehensive analysis of remote proctoring solutions: Features, limitations, and selection criteria," *Journal of Educational Computing Research*, vol. 60, no. 3, pp. 567-589, Jun. 2022.

[12] Y. Li, X. Wang, and H. Chen, "Privacy concerns in AI-based proctoring systems: A systematic analysis of risks and mitigation strategies," *IEEE Security & Privacy*, vol. 21, no. 1, pp. 45-53, Jan. 2023.

[13] R. Mehta, P. Patel, and S. Shah, "Object detection for proctoring using YOLOv4: Real-time identification of prohibited items," *IEEE International Conference on Machine Learning and Applications*, pp. 678-689, Dec. 2022.

[14] T. Nguyen, P. Le, and Q. Tran, "Survey of cheating detection methods in online examinations: From traditional to AI-based approaches," *ACM Computing Surveys*, vol. 55, no. 5, pp. 1-34, Jan. 2023.

[15] T. Okada, K. Yamamoto, and S. Tanaka, "Impact of cognitive load on cheating behavior in online examinations: A psychophysiological study," *Cognitive Science*, vol. 46, no. 3, pp. 78-92, Mar. 2022.

[16] R. Picard, E. Szu, and J. Kim, "Affective computing for academic integrity: Detecting stress and deception in remote assessments," *IEEE Transactions on Affective Computing*, vol. 12, no. 4, pp. 890-902, Oct. 2021.

[17] L. Qiu, Z. Wu, and Y. Liu, "Real-time behavioral analysis for online proctoring: A multimodal deep learning approach," *IEEE Transactions on Multimedia*, vol. 25, pp. 1234-1247, Mar. 2023.

[18] M. Rodriguez, C. Garcia, and L. Martinez, "Evolution of online proctoring technologies: From manual supervision to AI-powered systems," *Educational Technology & Society*, vol. 25, no. 3, pp. 112-125, Jul. 2022.

[19] P. Sharma, V. Singh, and R. Kumar, "Systematic review of artificial intelligence in online proctoring: Methods, datasets, and future directions," *Artificial Intelligence Review*, vol. 56, pp. 234-256, Feb. 2023.

[20] Talview Research, "State of online proctoring 2023: Global trends, challenges, and innovations," *Talview Technical Report*, pp. 1-45, Jan. 2023.

[21] S. Usmani, S. Alam, and M. Khan, "Comprehensive survey of computer vision techniques in online proctoring systems," *IEEE Access*, vol. 11, pp. 45678-45695, Mar. 2023.

[22] V. Venkatesh, K. Reddy, and P. Kumar, "Deep learning for facial behavior analysis in proctoring applications," *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 4, no. 2, pp. 234-248, Apr. 2022.

[23] X. Wang, Y. Zhang, and L. Sun, "Survey of gaze tracking applications in education: From attention monitoring to cheating detection," *ACM Transactions on Multimedia Computing, Communications, and Applications*, vol. 19, no. 3, pp. 1-22, Jan. 2023.

[24] J. Xu, W. Li, and F. Wang, "Real-time cheating detection using multimodal behavioral data and deep learning," *IEEE International Conference on Data Mining*, pp. 567-578, Dec. 2023.

[25] A. Yadav, S. Mishra, and R. Singh, "Comprehensive framework for online examination integrity: Integrating behavioral, cognitive, and environmental factors," *Journal of Educational Technology Systems*, vol. 52, no. 2, pp. 189-208, Jan. 2024.

[26] H. Zhao, M. Liu, and J. Chen, "Attention mechanisms for behavioral pattern recognition in online proctoring," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 6, pp. 789-802, Jun. 2023.

[27] V. Raghav, S. Kumar, and A. Singh, "Head pose and gaze tracking for online proctoring using OpenCV," *International Conference on Computer Vision*, pp. 156-167, Dec. 2021.

[28] A. Kumar, P. Sharma, and R. Singh, "Keystroke dynamics for continuous authentication in online examinations," *IEEE Transactions on Information Forensics*, vol. 17, pp. 2345-2358, Mar. 2022.

[29] M. Hasan, S. Ahmed, and R. Islam, "Hybrid CNN-LSTM architecture for cheating behavior detection," *IEEE Access*, vol. 11, pp. 12345-12358, Feb. 2023.

[30] Google Research, "MediaPipe: A framework for building multimodal applied ML pipelines," *Google Technical Report*, Jan. 2023.

[31] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

[32] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735-1780, Nov. 1997.

[33] D. Kingma and J. Ba, "Adam: A method for stochastic optimization," *International Conference on Learning Representations*, May 2014.

[34] A. Vaswani, N. Shazeer, and N. Parmar, "Attention is all you need," *Advances in Neural Information Processing Systems*, vol. 30, pp. 5998-6008, Dec. 2017.

[35] Y. LeCun, L. Bottou, and Y. Bengio, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278-2324, Nov. 1998.

[36] J. Redmon, S. Divvala, and R. Girshick, "You only look once: Unified, real-time object detection," *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 779-788, Jun. 2016.

[37] F. Zhang, V. Bazarevsky, and A. Vakunov, "MediaPipe face detection: A real-time face detection pipeline," *Google Research Technical Report*, Jun. 2020.

[38] L. Chen, J. Zhang, and W. Wang, "Behavioral biometrics for continuous authentication: A survey," *ACM Computing Surveys*, vol. 53, no. 4, pp. 1-36, Sep. 2020.

[39] Q. Wang, Y. Zhang, and X. Li, "Privacy-preserving behavioral analysis for online proctoring systems," *IEEE Transactions on Information Forensics and Security*, vol. 17, pp. 234-248, Jan. 2022.

[40] Y. Zhang, M. Chen, and L. Wang, "Federated learning for privacy-preserving AI in education: Challenges and opportunities," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 2, pp. 567-579, Feb. 2023.

[41] M. Abadi, A. Chu, and I. Goodfellow, "Deep learning with differential privacy," *ACM Conference on Computer and Communications Security*, pp. 308-318, Oct. 2016.

[42] M. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you? Explaining the predictions of any classifier," *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135-1144, Aug. 2016.

[43] S. Lundberg and S. Lee, "A unified approach to interpreting model predictions," *Advances in Neural Information Processing Systems*, vol. 30, pp. 4765-4774, Dec. 2017.

[44] S. Ge, Z. Liu, and Y. Li, "Eye tracking for gaze estimation: A survey of methods and applications," *IEEE Transactions on Visualization and Computer Graphics*, vol. 24, no. 4, pp. 1456-1467, Apr. 2018.

[45] Y. Chen, J. Wang, and S. Chen, "Learning gaze estimation from synthetic data," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 5, pp. 1234-1246, May 2019.

[46] Z. Liu, Y. Luo, and J. Wang, "Face detection and alignment with deep learning: A comprehensive survey," *IEEE Transactions on Image Processing*, vol. 30, pp. 234-248, Feb. 2021.

[47] M. Ahmed, A. Mahmood, and R. Islam, "Mouse dynamics for behavioral analysis: A comprehensive review," *IEEE Access*, vol. 9, pp. 45678-45690, Mar. 2021.

[48] Y. Liang, S. Samtani, and B. Guo, "Continuous authentication using behavioral biometrics: A systematic review," *IEEE Transactions on Dependable and Secure Computing*, vol. 18, no. 3, pp. 1456-1468, May 2021.

[49] M. Monaro, L. Gamberini, and G. Sartori, "Detecting fraudulent identity using keystroke dynamics: A machine learning approach," *IEEE Transactions on Information Forensics and Security*, vol. 13, no. 5, pp. 1234-1245, May 2018.

[50] K. He, X. Zhang, and S. Ren, "Deep residual learning for image recognition," *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770-778, Jun. 2016.

[51] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *International Conference on Learning Representations*, Apr. 2014.

[52] C. Szegedy, W. Liu, and Y. Jia, "Going deeper with convolutions," *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1-9, Jun. 2015.

[53] G. Huang, Z. Liu, and L. Van Der Maaten, "Densely connected convolutional networks," *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 4700-4708, Jul. 2017.

[54] S. Ren, K. He, and R. Girshick, "Faster R-CNN: Towards real-time object detection with region proposal networks," *Advances in Neural Information Processing Systems*, vol. 28, pp. 91-99, Dec. 2015.

[55] J. Carreira and A. Zisserman, "Quo vadis, action recognition? A new model and the kinetics dataset," *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 6299-6308, Jul. 2017.

[56] D. Tran, L. Bourdev, and R. Fergus, "Learning spatiotemporal features with 3D convolutional networks," *IEEE International Conference on Computer Vision*, pp. 4489-4497, Dec. 2015.

[57] C. Feichtenhofer, A. Pinz, and A. Zisserman, "Convolutional two-stream network fusion for video action recognition," *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1933-1941, Jun. 2016.

[58] L. Wang, Y. Xiong, and Z. Wang, "Temporal segment networks: Towards good practices for deep action recognition," *European Conference on Computer Vision*, pp. 20-36, Oct. 2016.

[59] G. Varol, I. Laptev, and C. Schmid, "Long-term temporal convolutions for action recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 40, no. 6, pp. 1510-1517, Jun. 2017.

[60] B. Zhou, A. Andonian, and A. Oliva, "Temporal relational reasoning in videos," *European Conference on Computer Vision*, pp. 803-818, Sep. 2018.

[61] C. Li, Q. Zhong, and D. Xie, "Collaborative spatiotemporal feature learning for video action recognition," *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 7872-7881, Jun. 2019.

[62] J. Duan, S. Zhou, and J. Wan, "CenterNet: Keypoint triplets for object detection," *IEEE International Conference on Computer Vision*, pp. 6569-6578, Oct. 2020.

[63] Z. Tian, C. Shen, and H. Chen, "FCOS: Fully convolutional one-stage object detection," *IEEE International Conference on Computer Vision*, pp. 9627-9636, Oct. 2019.

[64] M. Tan, R. Pang, and Q. Le, "EfficientDet: Scalable and efficient object detection," *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 10781-10790, Jun. 2020.

[65] Y. Kartynnik, A. Ablavatski, and I. Grishchenko, "Real-time facial surface geometry estimation," *Computer Vision Foundation*, Sep. 2018.

[66] Z. Chen, X. Wang, and L. Zhang, "Gaze tracking with consumer-grade cameras: Challenges and solutions," *IEEE Transactions on Multimedia*, vol. 23, pp. 234-245, Jan. 2021.

[67] M. Ho, T. Li, and J. Liu, "Multiperson gaze tracking for group behavior analysis," *IEEE International Conference on Computer Vision*, pp. 3456-3465, Oct. 2019.

[68] X. Zhang, Y. Sugano, and M. Fritz, "Learning appearance-based gaze estimation: A review of methods and datasets," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 3, pp. 678-691, Mar. 2019.

[69] N. Papernot, M. Abadi, and U. Erlingsson, "Scalable private learning with PATE," *International Conference on Learning Representations*, Apr. 2018.

[70] M. Buhrmester, D. Munch, and M. Arens, "Analysis of explainable artificial intelligence approaches: A systematic review," *ACM Computing Surveys*, vol. 52, no. 5, pp. 1-35, Oct. 2019.